

Long-Context

Alexandra Birch



Motivation

Long Documents (Code, legal, agent interaction histories)

Inference-time Reasoning



Review this paper [paper] with these guidelines [reviewer guidelines]



Summary

This paper presents a method for extending the context window of LLaMA 2 models from 4K to 32K tokens through continual pretraining....

Strengths

Comprehensive Evaluation: The paper provides extensive empirical evaluation across multiple dimensions...

Practical Impact: The models maintain or improve performance on short-context tasks while gaining long-context capabilities, addressing a key limitation of prior work (e.g., Chen et al., 2023) ...



If the entire world were made of glass marbles, could a human walk on the surface



This is a fascinating physics question! Let me think through what would happen.

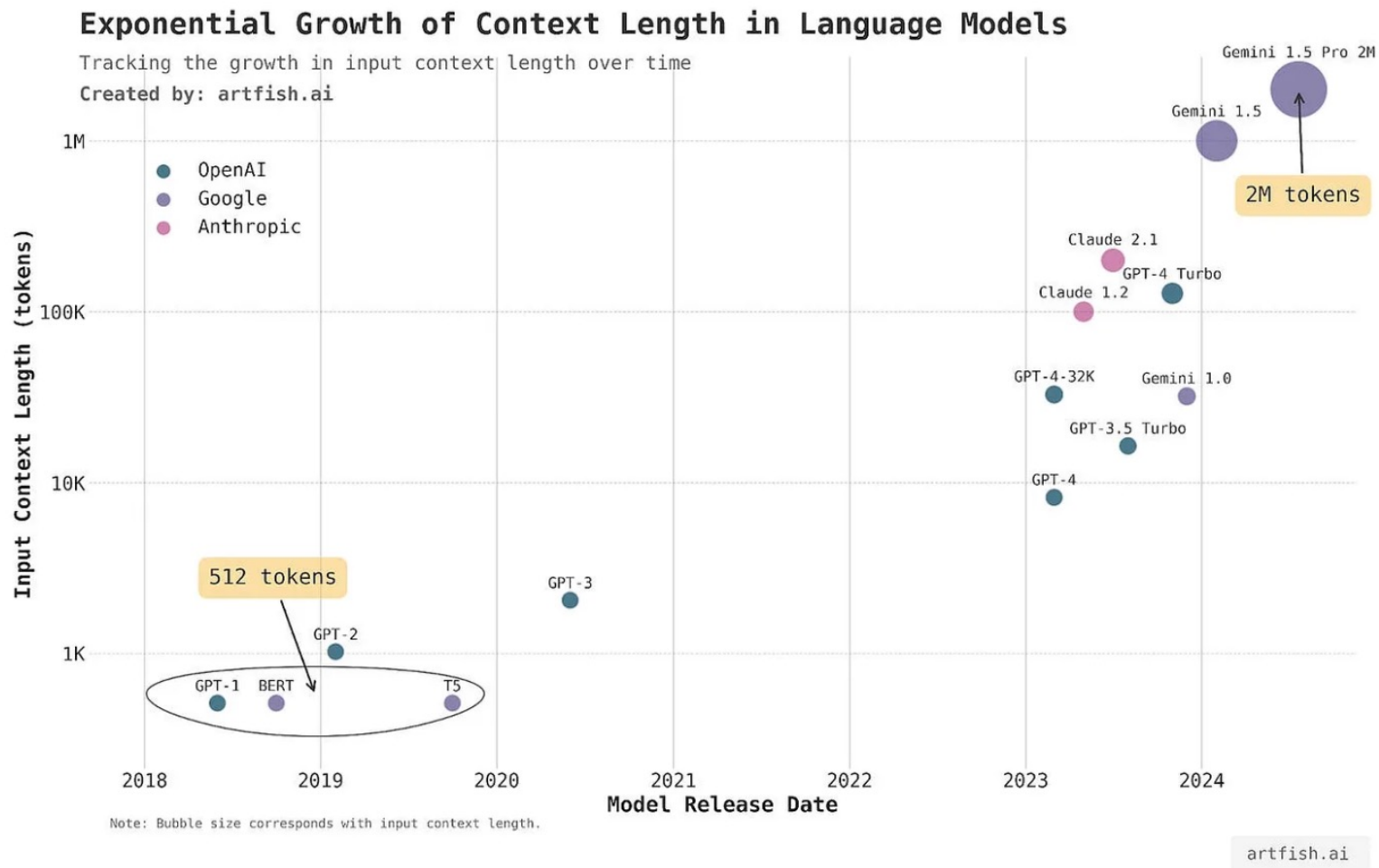
Short answer: No, a human could not walk on the surface - they would sink.

Here's why:

The Marble Packing Problem

Glass marbles can't pack perfectly. Even in the most efficient arrangement (face-centered cubic), spheres only fill about 74% of space....

Context Length



From [Medium/data-science](https://medium.com/data-science)

Current Frontier Models

Model family	Typical max input context window (tokens)	Notes
Anthropic Claude 4	200k standard; 500k enterprise; 1M beta tiers	May 2025
Google Gemini 3 Pro	1M tokens standard; up to 2M in some enterprise configs	Nov 2025: Longest context window of any large-scale foundation model
DeepSeek-R1	128k input tokens, 32k output	Jan 2025: Separate input and output context limits to support reasoning
OpenAI GPT-5	400k tokens input, 128k output	August 2025

Long-Context Modelling

Process, retrieve, and reason over **very long input and/or output sequences**, ranging **to millions of tokens**

Without long-context modeling, systems need **retrieval, truncation, or summarization**, which can lose critical information.

Core Challenges

Training Data Mismatch

- Most pretraining data consists of **short documents**.
- Training on long documents **extremely expensive** and **lack of realistic data**
- Models extrapolating to long contexts face:
 - Distribution shift
 - Positional generalization failures

Computational Cost

- Self-attention scales quadratically with sequence length
- This makes naive scaling prohibitively expensive in compute and memory.

Task Difficulty

- Even when long contexts are supported, models may:
 - Fail to retrieve relevant information
 - Overweight irrelevant or recent content
- Long context $\neq$ effective long-range reasoning.

Outline

- Positional Encodings
- Efficient Attention
- Evaluation
- Analysis
- Summary

Positional Encodings

Positional Encodings: Recap

Attention Is All You Need/Transformer

- Order of elements is crucial – word embeddings do not contain position
- Transformer processes all tokens in parallel – unlike recurrent models – no concept of sequential structure of data
- Token embeddings x_{pos} + Positional encodings PE_{pos} get concatenated before being passed to Transformer attention layers

$$z_{pos} = x_{pos} + PE_{pos}$$

Positional Encodings

Q: Why not use integer positional encodings?

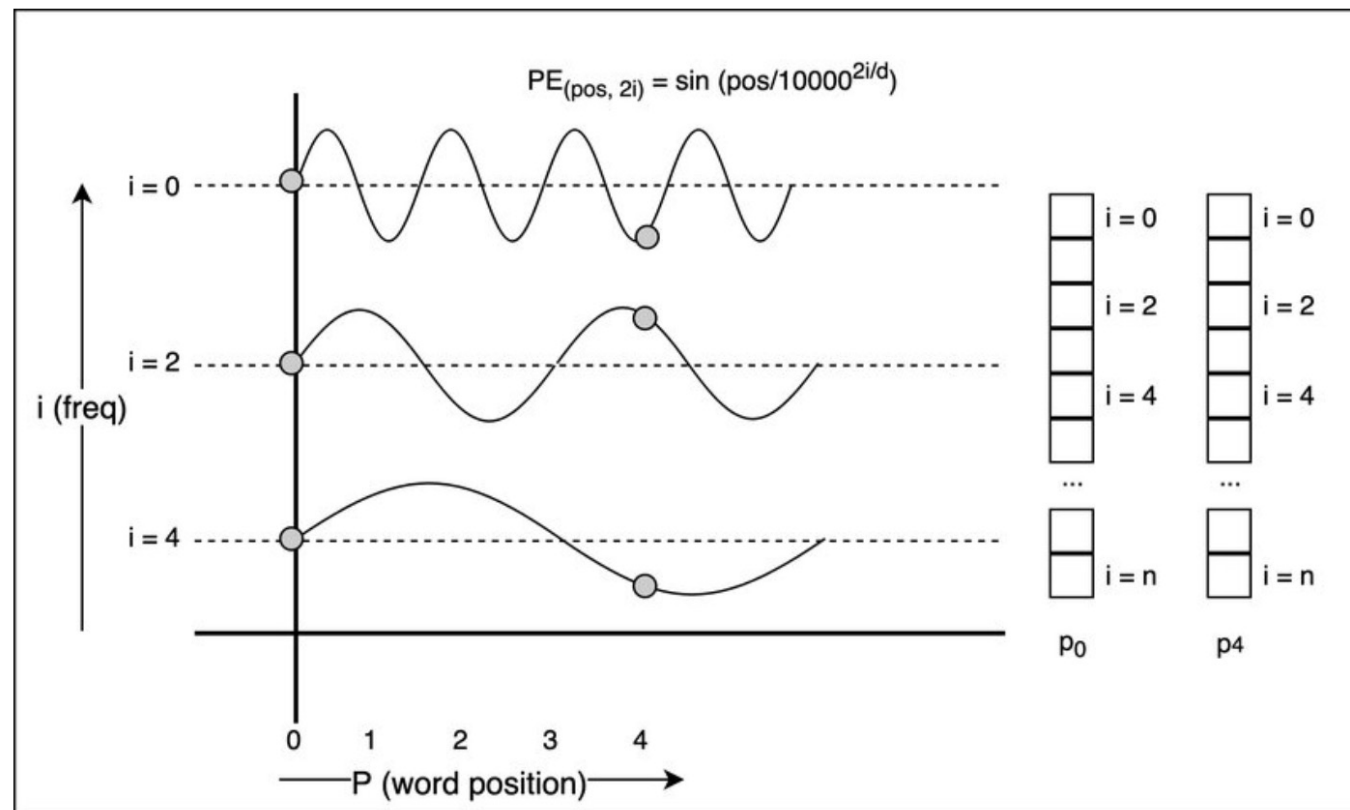
- Unbounded values:
 - Position 1000 is 1000× larger than position 1
 - Dominates the embedding magnitudes
 - Breaks numerical stability
- No generalization:
 - If trained on sequences up to length 512 Never seen positions 513, 514, ...
 - Can't handle longer sequences
- Poor scaling:
 - Position differences aren't meaningful Is position 100→101 similar to 500→501?
 - Hard for model to learn

Positional Encodings: Sinosoidal

Use multiple sine and cosine functions at different frequencies

Sinusoidal positional encodings are fixed

- Low frequency (high i) $\rightarrow$ slow oscillations (global position)
- High frequency (low i) $\rightarrow$ fast oscillations (local position)



[Researchgate.net](https://www.researchgate.net)

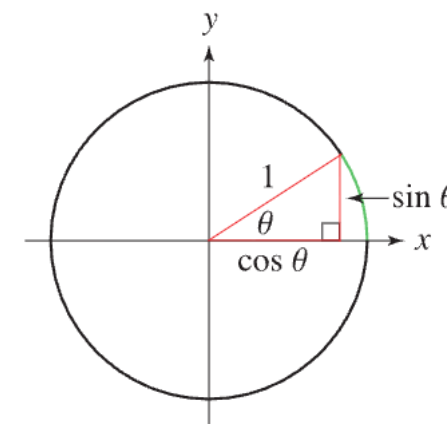
Positional Encodings: Sinosoidal

$$\text{PE}_{\text{pos}} = [\sin(\omega_0 \text{pos}), \cos(\omega_0 \text{pos}), \sin(\omega_1 \text{pos}), \cos(\omega_1 \text{pos}), \dots, \sin(\omega_{d/2-1} \text{pos}), \cos(\omega_{d/2-1} \text{pos})]$$

Where $\omega_i = 1/10000^{2i/d}$ frequency determines the speed of oscillation
 d = transformer embedding dim

- **Properties**

- sin cos pair defines a **single angle** on a unit circle
- PE sin cos vector forms a **unique fingerprint** for each position
- Can infer how far apart two tokens or pos are: dot product of two PE contain **relative distance**
- Deterministic (no learning)
- **Extrapolate** to longer sequences than seen in training



Positional Encodings: Sinosoidal

Let's see positions 0, 1, 2 with $d=4$ dimensions:

python

Simplified (wavelengths 1, 100)

```
pos=0: [sin(0/1), cos(0/1), sin(0/100), cos(0/100)]  
      = [0,          1,          0,          1      ]
```

```
pos=1: [sin(1/1), cos(1/1), sin(1/100), cos(1/100)]  
      ≈ [0.841,   0.540,   0.010,   0.9999   ]
```

```
pos=2: [sin(2/1), cos(2/1), sin(2/100), cos(2/100)]  
      ≈ [0.909,  -0.416,   0.020,   0.9998   ]
```

Fast dimensions (0-1) change a lot between positions

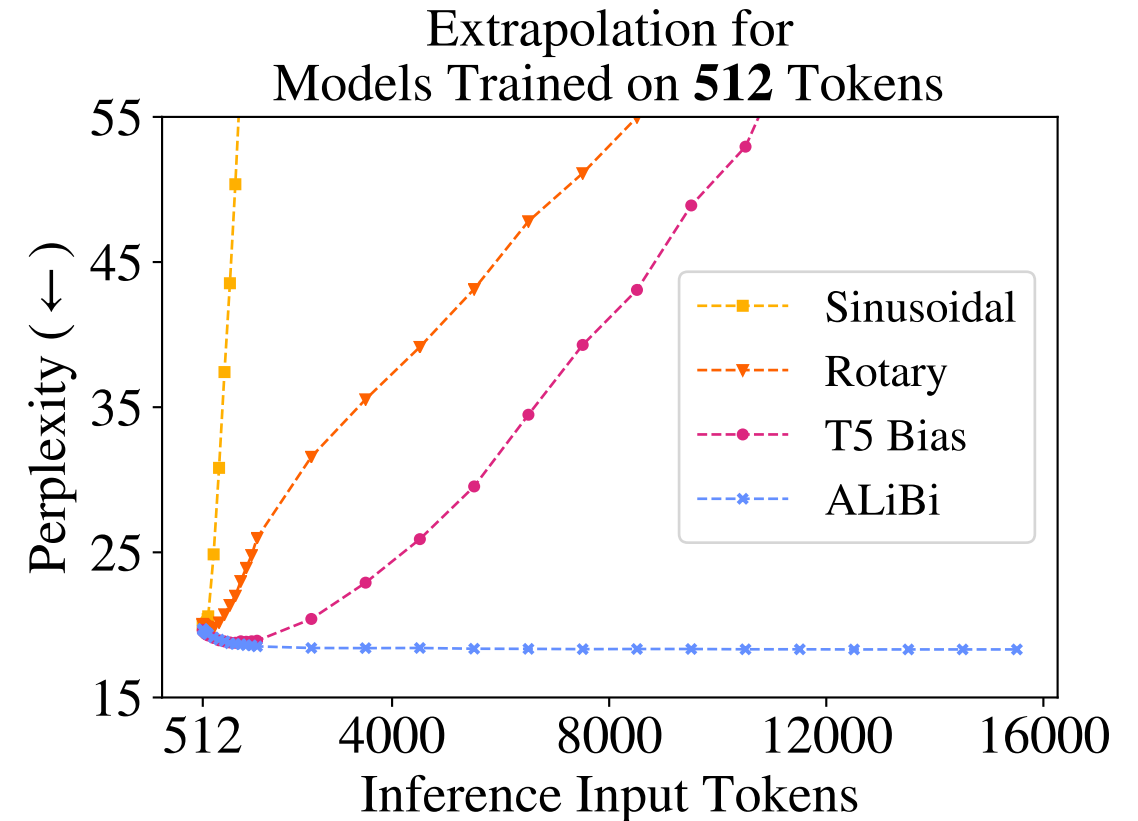
Slow dimensions (2-3) change very slowly

Together they uniquely identify each position

Positional Encoding Design

Sinusoidal PE does not extrapolate well beyond training length

Rotary methods are more promising



Press et al. (2021) [Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation](#)

RoPE

Addition (sinusoidal PE)

- Position is *attached* to the vector
- Relative distance must be *learned*

Rotation (RoPE)

- Position *transforms* the vector
- Turns absolute position → angle
- Turns relative position → angle difference so it is **built into the math** no learning required.

Su et al. (2021) [Roformer: Enhanced transformer with rotary position embedding](#)

RoPE

2D Example:

Query $q = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, **Key** $k = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Attention without position: $q \cdot k = 1$

Rotation matrix R for position pos : $\begin{bmatrix} \cos(pos) & -\sin(pos) \\ \sin(pos) & \cos(pos) \end{bmatrix}$

Query at position 1: $q_1 = R(1)q = \begin{bmatrix} \cos 1 \\ \sin 1 \end{bmatrix} \approx \begin{bmatrix} 0.54 \\ 0.84 \end{bmatrix}$ **Rotary position embedding**

Key at position 3: $k_3 = R(3)k = \begin{bmatrix} \cos 3 \\ \sin 3 \end{bmatrix} \approx \begin{bmatrix} -0.99 \\ 0.14 \end{bmatrix}$

Attention with position: $q_1 \cdot k_3 = (0.54)(-0.99) + (0.84)(0.14) = -0.41$

Now, relative position difference is 2. What is $\cos 2$? -0.41 !

ie. Attention sees **cos(relative distance)**.

RoPE

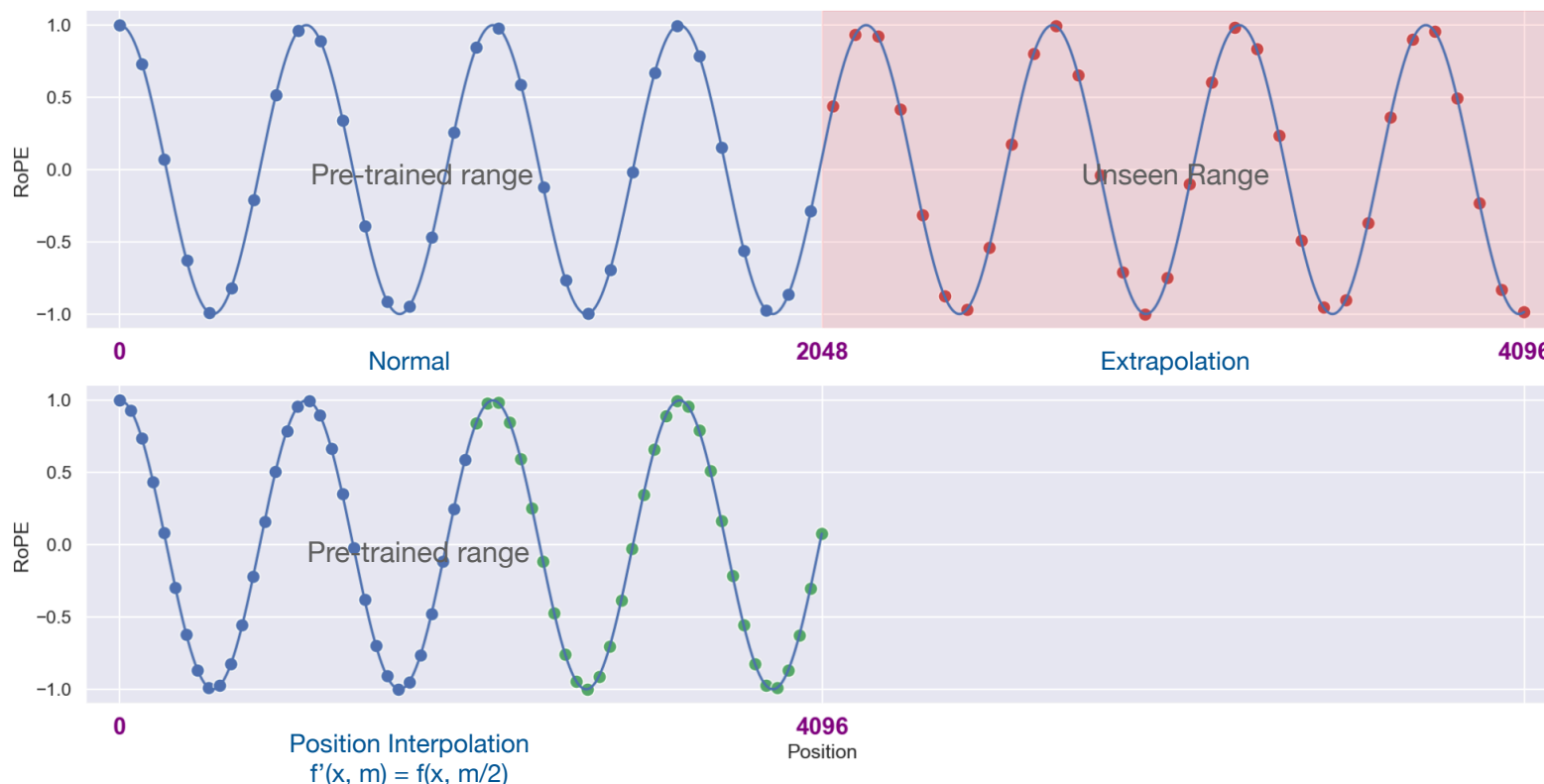
- Do not want to transform value V, only attention (thru Q and K)
- Rotation with **sin** and **cos** form an orthogonal transform which is
 - Length-preserving
 - Direction-sensitive
- This gives:
 - Stability
 - Extrapolation
 - No distortion of content magnitude

Model	Validation	Test
BERT-512	64.13%	67.77%
WoBERT-512	64.07%	68.10%
RoFormer-512	64.13%	68.29%
RoFormer-1024	66.07%	69.79%

Extending context to 1024 shows RoPE improves on “long” document understanding

Extending Context Beyond Training

- Training with long context is expensive!
- And RoPE struggles to generalize past the context window seen during pre-training. Why?
High-frequency rotations wrap around too fast and relative distances become noisy



Performance degrades on unseen range

Position Interpolation:
Downscale position indices to reside in pretrained range
Extended from 2048 to 32k
Fine tuning on only 1000 steps
Negligible compared to pre-train

Chen et al. (2023) [Extending Context Window of Large Language Models via Positional Interpolation](#)

NTK Scaling

Problem! With Position Interpolation when you compress positions linearly, adjacent tokens become harder to distinguish.

Eg. Original positions: 100, 101, 102, 103

After interpolation 4 x compression: 25.0, 25.25, 25.5, 25.75

NTK (Neural Tangent Kernel – why scale changes learning dynamics – just theoretical connection)

Scaling behaviour of attention – want it to be stable as context length increases

- RoPE rotation angle $\theta = \omega_i \cdot pos$ remember ω_i is angular frequency: Large – rotate fast, small - slow
- RoPE-NTK $\theta = (\alpha \omega_i) \cdot pos$ where $\alpha < 1$ for longer contexts

Position Interpolation: Compress position indices, keeping frequencies fixed

NTK Scaling: Keep position indices, adjust rotation frequencies

Optional reading <https://amaarora.github.io/posts/2025-09-21-rope-context-extension.html>

YaRN

YaRN: Yet another RoPE extension method: Position Interpolation in parts

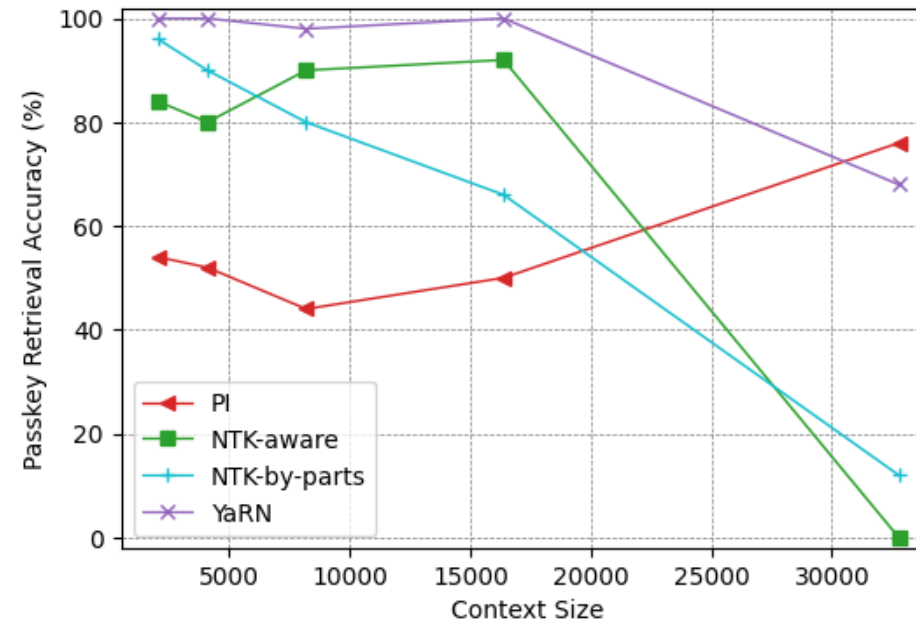
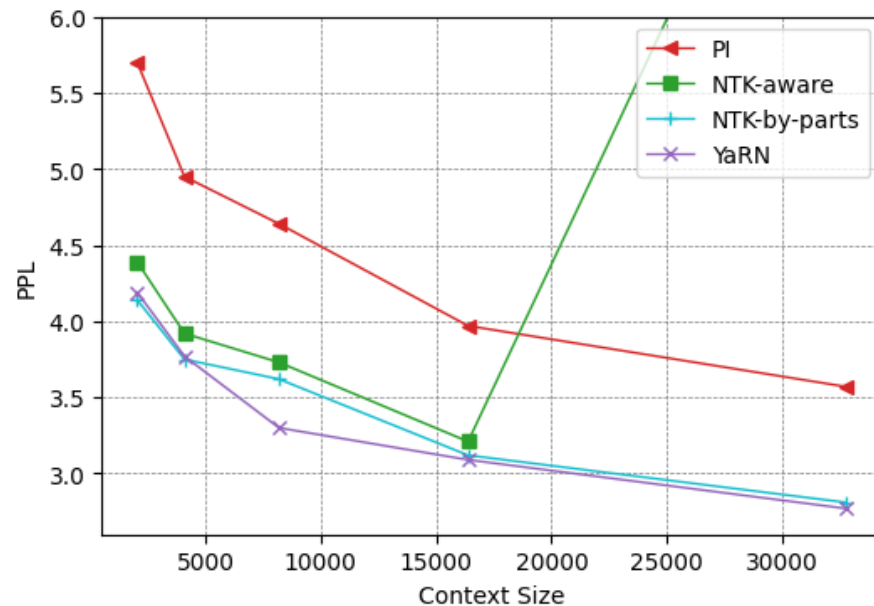
Pretrained Context length = L

Wavelength λ = number of tokens for a RoPE dimension to complete rotation

$\ll L$ no positional interpolation

in between some positional interpolation

$\geq L$ positional interpolation



Pushed context > 128k

Peng et al. (2023) [YaRN: Efficient Context Window Extension of Large Language Models](#)

Efficient Attention

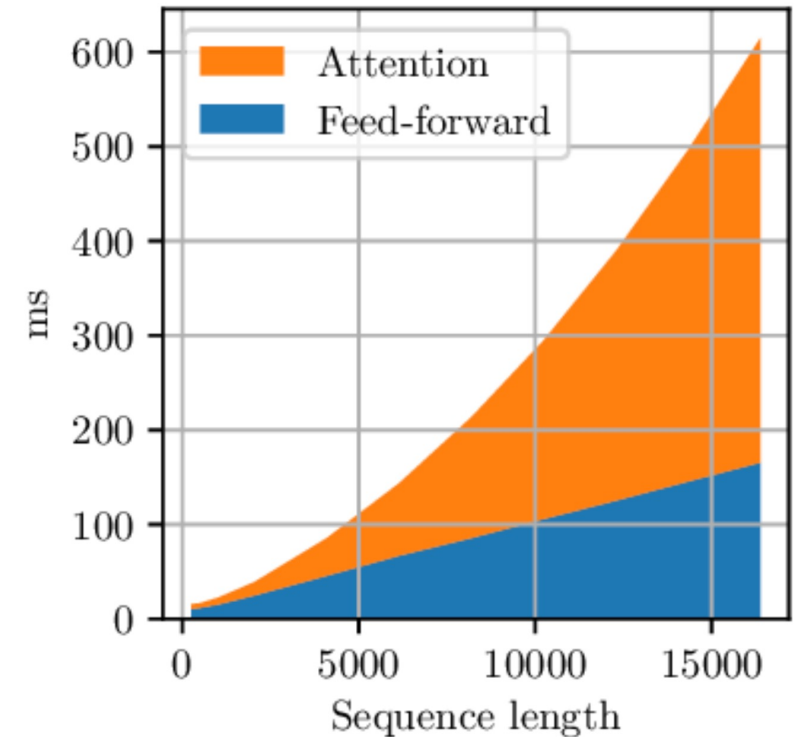
Complexity

The Quadratic Bottleneck in Transformers

Given a sequence of length n , and matrices Query Q , Key K , Value V of dimension d :

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V$$

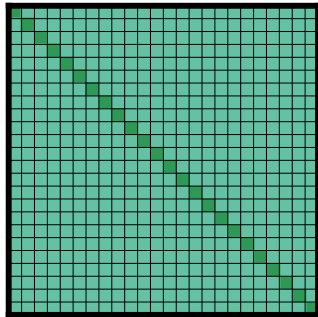
Where QK^T results in an $n \times n$ matrix, leading to $O(n^2)$ complexity



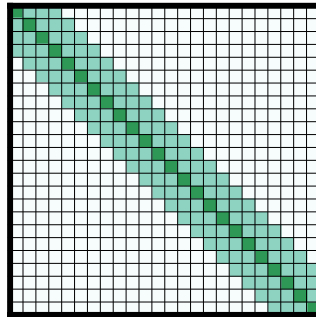
Sparse Attention

Sparse attention reduces complexity to $O(n)$: only compute attention weights for a subset of possible pairs

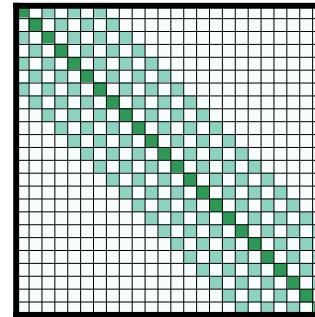
Beltagy et al. (2020) [Longformer: The Long-Document Transformer](#)



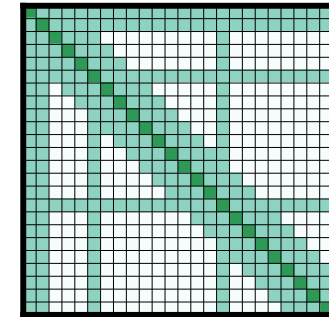
(a) Full n^2 attention



(b) Sliding window attention



(c) Dilated sliding window



(d) Global+sliding window

Inspired by locality of CNNs

Global: specific tokens attend to entire seq for local detail & global understanding

Showed could process sequences beyond 512 tokens – to 4096

Current Frontier Models

Model family	Typical max input context window (tokens)	Notes
Anthropic Claude 4	200k standard; 500k enterprise; 1M beta tiers	May 2025
Google Gemini 3 Pro	1M tokens standard; up to 2M in some enterprise configs	Nov 2025: Longest context window of any large-scale foundation model
DeepSeek-R1	128k input tokens, 32k output	Jan 2025: Separate input and output context limits to support reasoning
OpenAI GPT-5	400k tokens input, 128k output	August 2025

Really Long Context

- Rope scaling methods eg. YaRN
- Efficient attention

Other:

- Train on short sequences and then on very few long context examples – 0.1% of original pretraining data
- Don't recompute what you've already seen: KV caching and KV compression

However effectively using 100k+ attention remains challenging due to attention degradation over long distances

Evaluation

Needle in the Haystack

Can the model access long context? Simple, controllable sanity check

- The model is given a **very long context** (e.g. 50k–200k tokens).
Most of the text is **irrelevant filler** (the “haystack”): Eg. Wikipedia articles, News text
- Somewhere in the middle (often randomly placed) is a **single critical fact** (the “needle”).
- Ask the model to retrieve this statement
- Iterate over various document depths (where the needle is placed) and context lengths to measure performance

Needle in the Haystack

Input Context

Write a high-quality answer for the given question using only the provided search results (some of which might be irrelevant).

Document [1] (Title: Asian Americans in science and technology) Prize in physics for discovery of the subatomic particle J/ψ . Subrahmanyan Chandrasekhar shared...

Document [2] (Title: List of Nobel laureates in Physics) The first Nobel Prize in Physics was awarded in 1901 to Wilhelm Conrad Röntgen, of Germany, who received...

Document [3] (Title: Scientist) and pursued through a unique method, was essentially in place. Ramón y Cajal won the Nobel Prize in 1906 for his remarkable...

Question: who got the first nobel prize in physics

Answer:

Desired Answer

Wilhelm Conrad Röntgen

Needle in the Haystack

Input Context

Write a high-quality answer for the given question using only the provided search results (some of which might be irrelevant).

Document [1] (Title: List of Nobel laureates in Physics) ...

Document [2] (Title: Asian Americans in science and technology) ...

Document [3] (Title: Scientist) ...

Question: who got the first nobel prize in physics
Answer:

Desired Answer

Wilhelm Conrad Röntgen

Modulating position of needle

Input Context

Write a high-quality answer for the given question using only the provided search results (some of which might be irrelevant).

Document [1] (Title: Asian Americans in science and technology) ...

Document [2] (Title: List of Nobel laureates in Physics) ...

Document [3] (Title: Scientist) ...

Document [4] (Title: Norwegian Americans) ...

Document [5] (Title: Maria Goeppert Mayer) ...

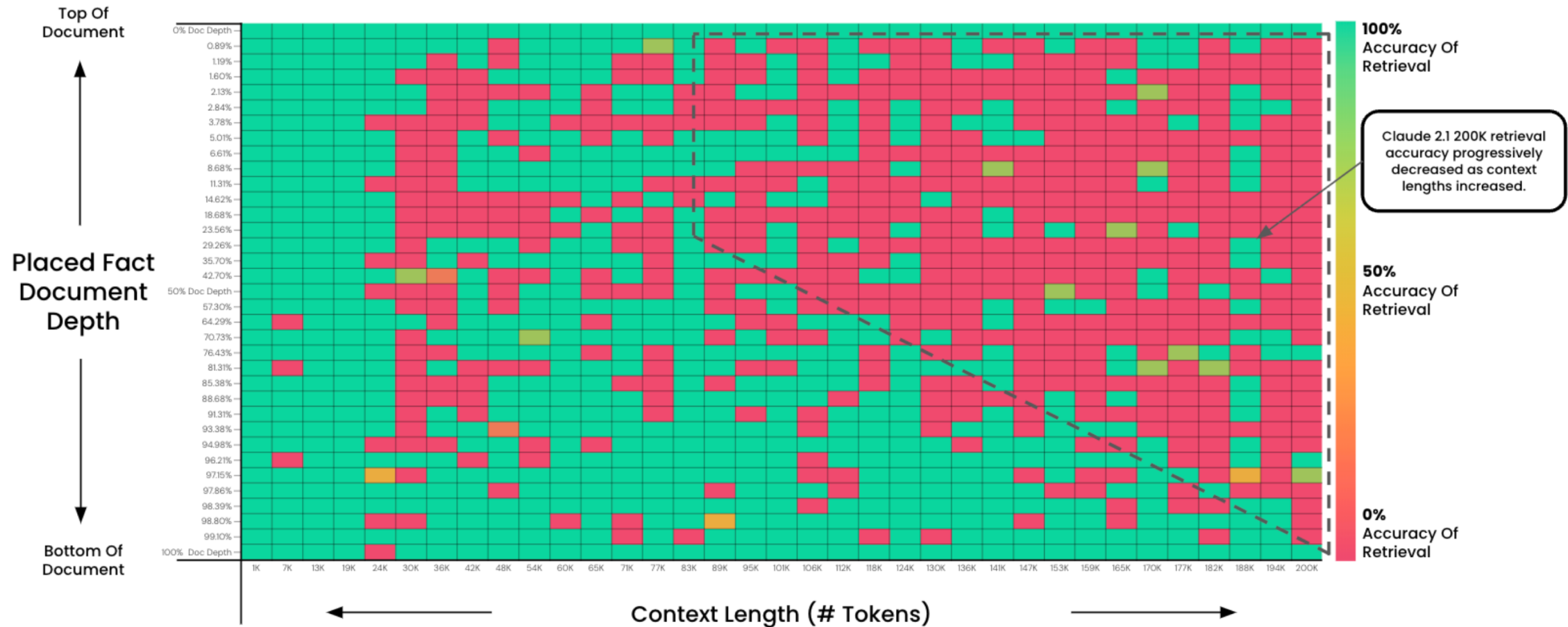
Question: who got the first nobel prize in physics
Answer:

Desired Answer

Wilhelm Conrad Röntgen

Modulating size haystack adding documents

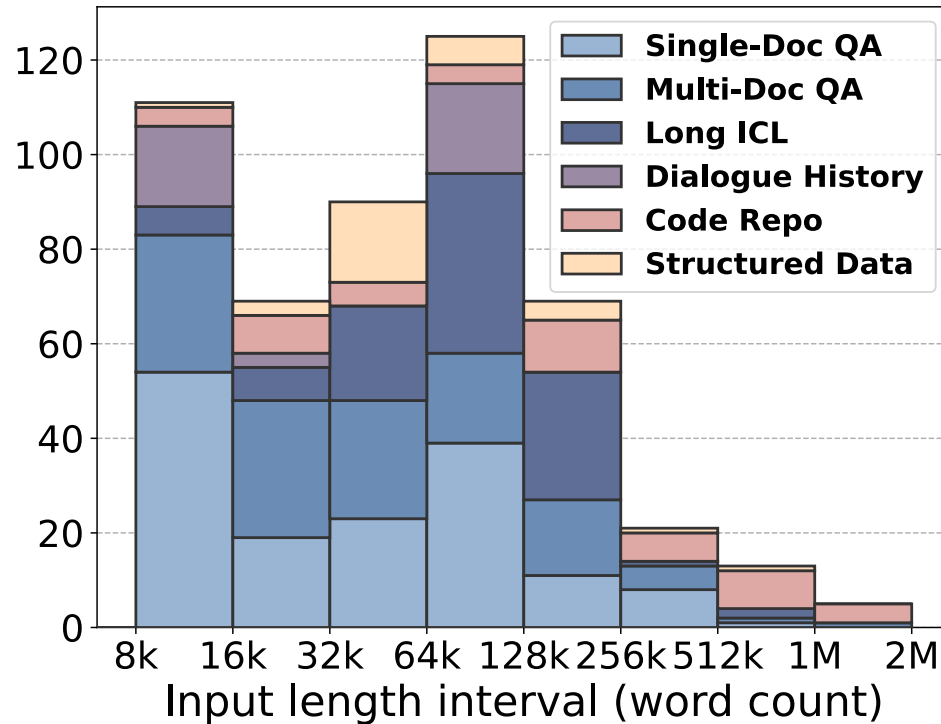
Needle in the Haystack



Claude 2.1: 0-200k token context
35 different document depths top > bottom
35 different context lengths left > right
Evaluated with GPT4 for accuracy

Kamradt et al. (2023) [Needle in a haystack – pressure testing llms](#)

LongBench



503 challenging MCQ
Context from 8k-2M words
6 major task categories
100 expert annotators
With automated and human review

Bai et al. (2023) [LongBench v2: Towards Deeper Understanding and Reasoning on Realistic Long-context Multitasks](#)

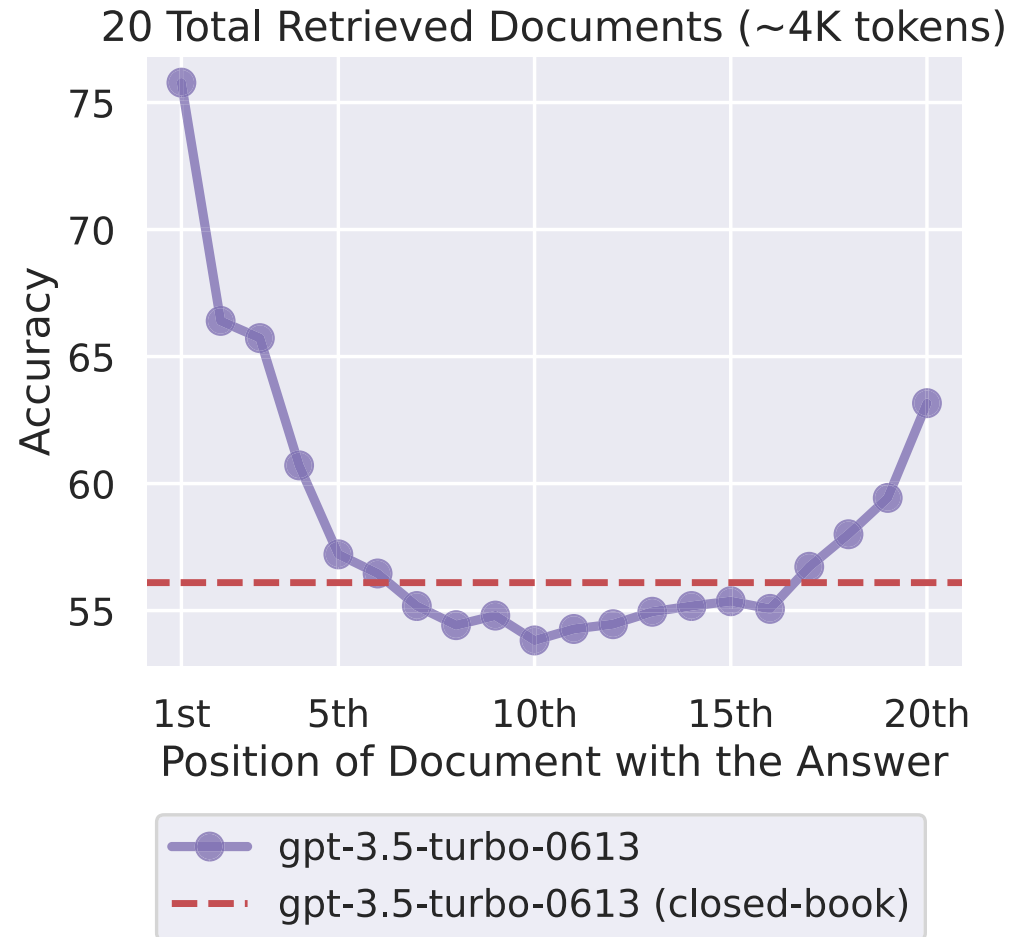
LongBench

Dataset	Source	#data	Length	Expert Acc	Expert Time*
<i>I. Single-Document QA</i>		175	51k	55%	8.9 min
Academic	Paper, textbook	44	14k	50%	7.3 min
Literary	Novel	30	72k	47%	8.5 min
Legal	Legal doc	19	15k	53%	13.1 min
Financial	Financial report	22	49k	59%	9.0 min
Governmental	Government report	18	20k	50%	9.5 min
Detective	Detective novel	22	70k	64%	9.3 min
Event ordering	Novel	20	96k	75%	9.4 min
<i>II. Multi-Document QA</i>		125	34k	36%	6.1 min
Academic	Papers, textbooks	50	27k	22%	6.1 min
Legal	Legal docs	14	28k	64%	8.8 min
Financial	Financial reports	15	129k	40%	7.0 min
Governmental	Government reports	23	89k	22%	6.0 min
Multi-news	News	23	15k	61%	5.3 min
<i>III. Long In-context Learning</i>		81	71k	63%	8.3 min
User guide QA	Electronic device, software, instrument	40	61k	63%	9.9 min
New language translation	Vocabulary book (<i>Kalamang, Zhuang</i>)	20	132k	75%	5.4 min
Many-shot learning	Multi-class classification task	21	71k	52%	8.0 min
<i>IV. Long-dialogue History Understanding</i>		39	25k	79%	8.2 min
Agent history QA	LLM agents conversation	20	13k	70%	8.3 min
Dialogue history QA	User-LLM conversation	19	77k	89%	6.5 min
<i>V. Code Repository Understanding</i>		50	167k	44%	6.4 min
Code repo QA	Code repository	50	167k	44%	6.4 min
<i>VI. Long Structured Data Understanding</i>		33	49k	73%	6.4 min
Table QA	Table	18	42k	61%	7.4 min
Knowledge graph reasoning	KG subgraph	15	52k	87%	6.2 min

Bai et al. (2023) [LongBench v2: Towards Deeper Understanding and Reasoning on Realistic Long-context Multitasks](#)

Analysis

Lost in the Middle



Attention $\neq$ reasoning.
Model can attend to
middle tokens but doesn't
effectively use them.

Liu et al. (2024) [Lost in the Middle: How Language Models Use Long Contexts](#), ACL

Summary

- Tricky and expensive to get long context tasks to work
- Positional Encodings are key to extending modelling long sequences
- Huge advances in the last couple of years but no free lunch

Eg. tradeoffs between efficiency vs quality of attention

- Not clear if using 1M tokens works as well as we think and very expensive
- Still unclear how long context compares to RAG, or context engineering